AAROGYASETU • आरोग्यसेतु

"Healthcare Beyond Boundaries" — Visual Project & Technical Development Blueprint

An installable, mobile-first Progressive Web App (PWA) designed for rural and remote communities with slow or intermittent 2G internet. Unites a role-separated **Patient Portal (11 screens)** and **Doctor Portal (9 screens)** with client-side image compression, explicit offline/retry states, consent-gated non-PHI SMS appointment updates, Smart OCR report verification, doctor-approved medicine reminders, tokenized QR profiles, non-diagnostic AI education, and Emergency SOS.

How to Use This Blueprint — Recommended 6-Stage Build Order for Beginners & Hackathon Teams

Stage 1 (Foundation & RBAC): Set up Python FastAPI + SQLite (`users`, `patient_profiles`, `doctor_profiles`) and JWT auth for `patient` vs `doctor` roles.
Stage 2 (Installable PWA Shell & Teal/Slate UI): Build the React + TypeScript app shell (`manifest.json` + `sw.js` < 85 KB) with the persistent 2G Data Saver bar, KB saved counter, and Demo Offline toggle.
Stage 3 (Appointments & SMS Updates Module): Implement doctor search, slot booking, status transitions (`requested` -> `confirmed` / `rescheduled` / `cancelled`), and the idempotent, zero-PHI SMS queue with Demo Log fallback.
Stage 4 (Low-Bandwidth Chat, Canvas Compression & Smart OCR): Add consultation messaging (`Saved/Pending/Synced/Failed+Retry`), client-side image compression (`3.2 MB -> 38 KB`), and OCR report extraction with mandatory Doctor Review.
Stage 5 (Clinical Notes, Medicine Reminders, QR & SOS): Add doctor clinical notes, doctor-approved medicine schedules, tokenized QR sharing, non-diagnostic AI Health Assistant, and manual-dialer Emergency SOS.
Stage 6 (Demo Verification): Seed fictional rural personas (`Ramesh Kumar` & `Dr. Priya Sharma`) and verify the 3-minute dual-portal walkthrough.

SECTION 1 | PROJECT OVERVIEW

Attribute	Specification
Project Name & Script	**AAROGYASETU** | Hindi Name: **आरोग्यसेतु** | Tagline: **"Healthcare Beyond Boundaries"**
Target Platform	Installable Mobile-First Progressive Web App (PWA) for Android/iOS phones & desktop browsers (Chrome/Edge).
Design Reference	Inspired by low_bandwidth_telemedicine_platform.tsx: Slate-900 header, Teal-600 cards, Data-Saved KB counter, 2G toggle & Amber offline banner.
Distinct User Roles	**1. Patient Portal** (Rural citizens & families) | **2. Doctor Portal** (PHC doctors & remote tele-clinicians).

1.1 The Rural Healthcare Access Problem & Proposed Solution

The Problem: Families in rural villages often travel 15–50 km to reach a Primary Health Centre (PHC) or specialist doctor, losing daily wages for simple consultations or lab report follow-ups. Existing telemedicine apps rely on high-bandwidth video calls and heavy uncompressed image uploads that fail on intermittent 2G/Edge village networks.

Proposed Solution: AAROGYASETU (आरोग्यसेतु) replaces fragile video streaming with an installable, offline-capable PWA that compresses medical report photos by ~98% in the browser before upload, queues messages and logs with explicit retry states, keeps patients updated via consent-gated non-PHI SMS notifications, and equips doctors with consultation-scoped triage, human-verified OCR report reading, and structured medicine schedule tools.

Intended User Group	Real-World Context & Constraints	Expected Benefits
1. Rural Patients & Families	Entry-level smartphones, slow/unreliable 2G connectivity, limited data packs, bilingual (English/Hindi) literacy needs.	• Book doctor consultations without long travel. • Upload compressed lab reports (~38 KB instead of 3.2 MB). • Receive SMS updates for confirmations & reschedules. • Access past records, doctor-approved medicine reminders & Emergency SOS offline.

Intended User Group	Real-World Context & Constraints	Expected Benefits
2. PHC & Remote Doctors	High patient volume across village clinics; need rapid triage, structured lab review, and clear follow-up tracking.	• Manage appointment requests (`Confirm`, `Reschedule`, `Cancel`) with automated patient SMS alerts. • Review OCR-extracted lab values side-by-side with the image. • Publish clinical notes & medicine schedules; scan temporary consented QR tokens.



Figure 1: AAROGYASETU (आरोग्यसेतु) — 8-Module Feature Overview & Clinical Safety Ecosystem.

SECTION 2 | SCOPE AND USER JOURNEYS (PATIENT & DOCTOR WORKFLOWS)

2.1 Patient Journey (Registration to Follow-Up)

- **1. Registration, Profile & Consent:** Patient installs or opens the PWA, registers (`role = patient`), records village, allergies, chronic conditions, and emergency contact (`P-02`), and explicitly opts into **SMS Appointment Alerts**.
- **2. Doctor Search & Appointment Booking:** Patient filters verified doctors by specialty and language (`P-04`), picks an available time slot (`P-05`), and submits a booking (`status = requested`). An idempotent, zero-PHI booking confirmation SMS is queued immediately.
- **3. Consultation Messaging & Compressed Report Upload:** Inside the appointment consultation (`P-06`, `P-07`), the patient exchanges low-data text messages (`Saved -> Pending -> Synced`) and selects a camera photo of a medical report. The browser compresses the image (`3.2 MB -> 38 KB`) before uploading for OCR extraction.
- **4. Records, Medicine Reminders, QR & SOS:** After the doctor reviews the report and signs the consultation note, the patient views their verified record (`P-07`), marks daily doctor-approved medicine doses `Taken` or `Skipped` (`P-09`), asks educational questions in the **AI Health Assistant (`P-08`)**, shares a temporary **Tokenized QR Code (`P-10`)** at a clinic, or uses **Emergency SOS (`P-11`)**.

2.2 Doctor Journey (Triage to Clinical Sign-Off)

- **1. Clinician Sign-In & Availability Setup:** Doctor signs into the dedicated Doctor Portal (`D-01`), sets specialty, registration number, spoken languages, and weekly consultation time windows (`D-02`).
- **2. Appointment Request Management & Status Updates:** From the dashboard (`D-03`) and queue (`D-04`), the doctor reviews `requested` appointments and clicks **Confirm**, **Reschedule** (proposing a new slot), or **Cancel**. Each status update triggers a consent-gated, non-PHI SMS notification to the patient.
- **3. Consultation-Scoped Patient Review & Messaging:** Doctor opens the assigned patient's profile (`D-05` — strictly restricted by RBAC to assigned consultations) and replies to the patient's messages (`D-06`).
- **4. Smart OCR Report Verification, Notes & Medicine Schedule:** Doctor inspects the uploaded report alongside machine-extracted OCR values (`D-07`), corrects any misread numbers, marks the report **Doctor Reviewed**, signs the clinical consultation note (`D-08`), and approves a **Medicine Reminder Schedule (`D-09`)** for follow-up tracking.

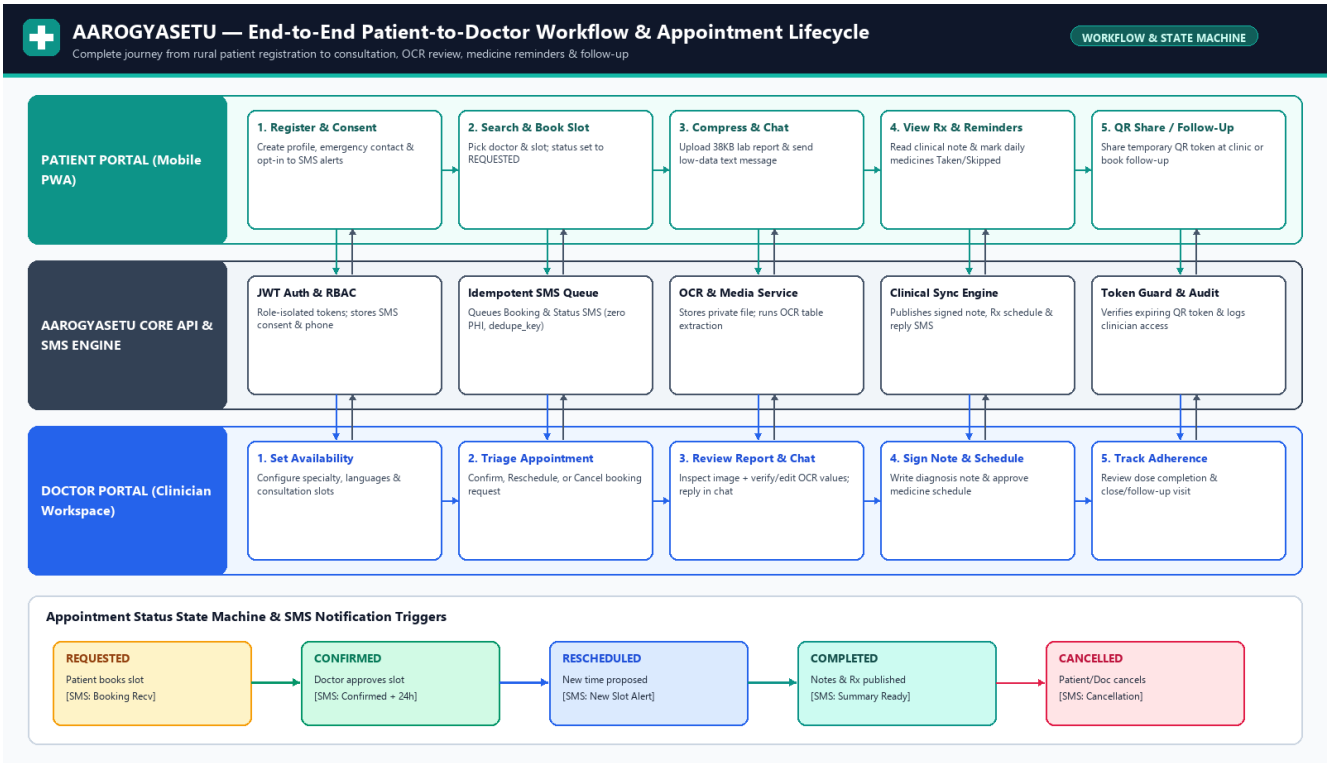


Figure 2: Complete Patient-to-Doctor Swimlane Journey & Appointment Status Lifecycle (`Requested -> Confirmed / Rescheduled / Cancelled -> Completed`).

SECTION 3 | APP INTERFACES — MOBILE-FIRST WIREFRAMES & SCREEN SPECIFICATIONS

Every screen uses a consistent low-bandwidth visual language inspired by low_bandwidth_telemedicine_platform.tsx: a Slate-900 telemetry header (`2G Saver: ON | Transferred KB | Data Saved KB`), high-contrast Teal-600/Blue-600 action cards, accessible labels, and explicit UI state handling (`Loading`, `Empty`, `Offline`, `Pending`, `Success`, `Error`).

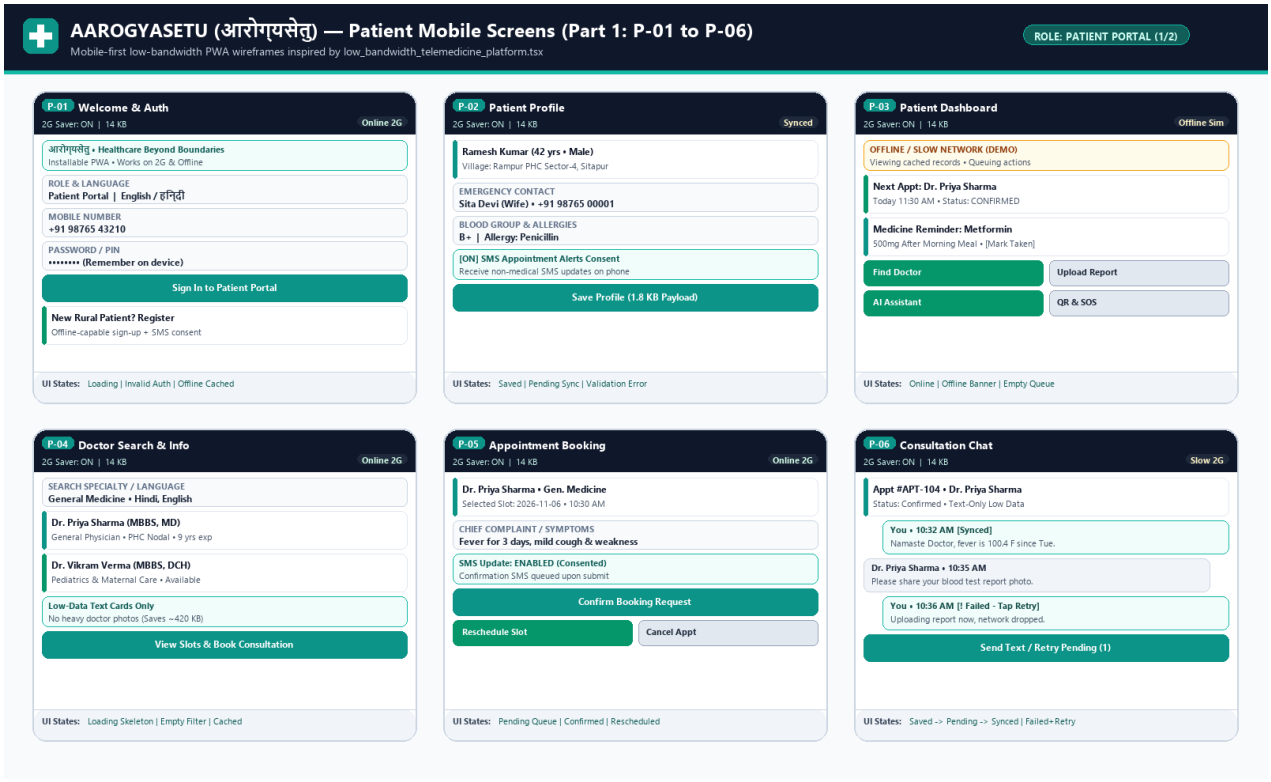


Figure 3A: Patient Portal Mobile Wireframes (Part 1: Screens P-01 to P-06).

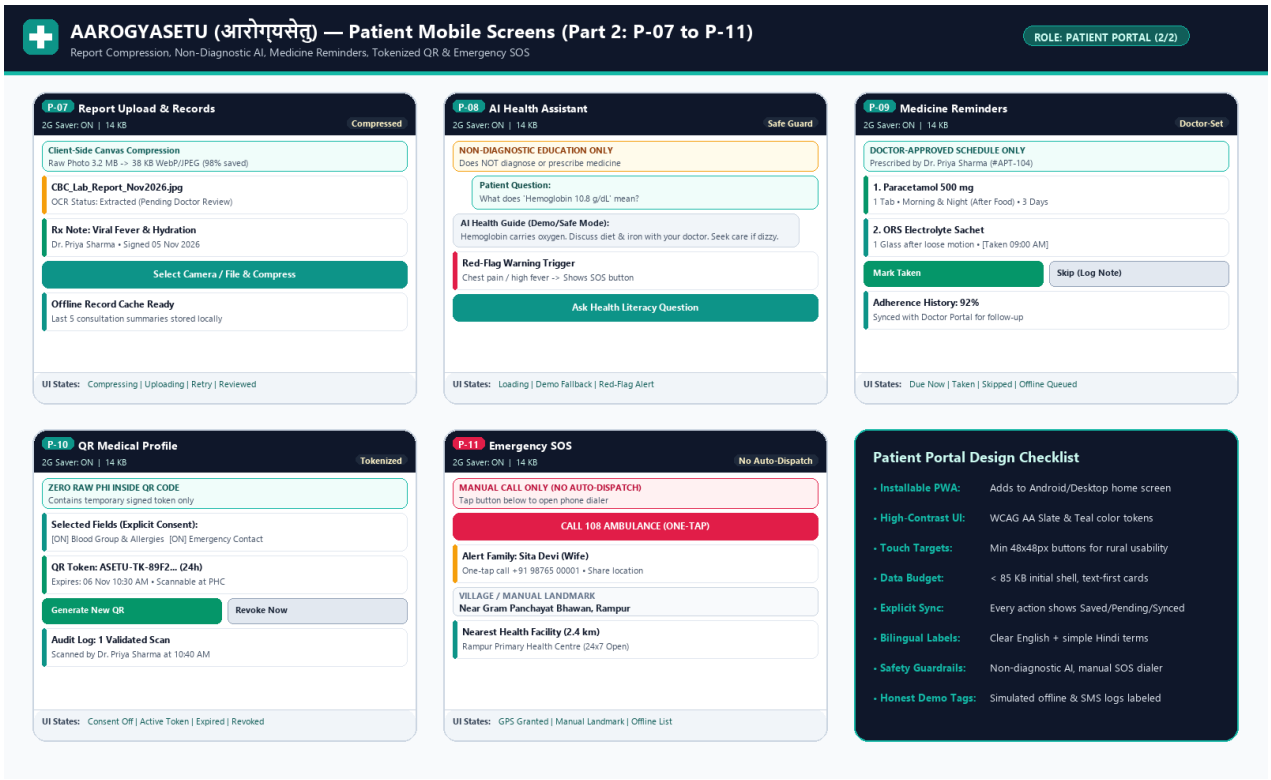


Figure 3B: Patient Portal Mobile Wireframes (Part 2: Screens P-07 to P-11 & Accessibility Checklist).

3.1 Complete Patient Portal Screen Specifications (11 Screens: P-01 to P-11)

Screen	Purpose	Main UI Elements	User Actions	States (Loading / Empty / Offline / Pending / Success / Error)
P-01 Welcome, Sign-Up & Sign-In	Role-aware entry point with PWA install prompt & fast login/registration.	• Brand banner (`AAROGYASETU / आरोग्यसेतु`) • Patient/Doctor role tabs • Phone/Email & Password inputs • `Install App` banner & Demo Quick-Fill	• Sign in or register as Patient • Install PWA to home screen • Fill demo credentials	• **Loading:** `Signing in...` spinner • **Empty:** Field required cues • **Offline:** Allows SOS access; notes auth requires connection • **Pending:** Token exchange • **Success:** Routes to `P-03` • **Error:** Invalid credentials retry
P-02 Patient Profile & Consent	Manage rural demographics, allergies, emergency contact & SMS consent.	• Name, Age/DOB, Gender, Village • Blood Group, Allergies, Conditions • Emergency Contact Name & Phone • **SMS Appointment Alerts Consent Toggle**	• Edit profile fields • Toggle SMS consent `ON`/`OFF` • Save (`1.8 KB` JSON)	• **Loading:** Input placeholders • **Empty:** Prompts emergency contact setup • **Offline:** `Saved Locally — Pending Sync` • **Pending:** `Saving 1.8 KB...` • **Success:** `Profile Synced` badge • **Error:** Phone format / Retry
P-03 Patient Dashboard	Low-bandwidth hub showing network banner, data saved, next visit & due meds.	• Amber **Offline Banner** (when offline) • `Data Saved KB` telemetry pill • Next Appointment status card • Due Medicine card + 2x2 action grid	• Open active consultation • Mark due dose `Taken` • Toggle `Simulate Offline (Demo)` • Launch any module	• **Loading:** Text-line skeleton • **Empty:** `Book First Consultation` CTA • **Offline:** Amber banner (`Viewing cached data`) • **Pending:** `1 Queued Item` • **Success:** `All Records Synced` • **Error:** `Tap to Retry Sync`
P-04 Doctor Search & Details	Browse doctors by specialty, language & schedule without heavy portraits.	• Specialty & Language filter chips • Text-first Doctor Cards (Name, Quals, PHC, Languages, Open Slots) • Expandable Doctor Details sheet	• Filter by specialty/language • View doctor qualifications & slots • Tap `Select Slot & Book`	• **Loading:** Card outlines • **Empty:** `No doctors match filter` • **Offline:** Displays cached directory • **Pending:** Refreshing slots • **Success:** Verified list rendered • **Error:** Uses cached list + Retry
P-05 Appointment Booking & Details	Book new slot or view, reschedule, or cancel an appointment.	• Doctor summary & 15m slot chips • Chief complaint input • SMS consent indicator • Status timeline + `Reschedule`/`Cancel`	• Submit booking (`requested`) • Reschedule to a new slot • Cancel appointment • View SMS alert log	• **Loading:** Fetching open slots • **Empty:** `No slots on selected date` • **Offline:** Saves booking draft locally • **Pending:** `Submitting request...` • **Success:** `#APT-104 Requested + SMS Queued` • **Error:** Slot conflict / Retry
P-06 Consultation Messaging	Low-bandwidth text chat between patient and assigned doctor.	• Appointment header (`#APT-104`) • Message bubbles + sync badges • Symptom quick-chips • Input bar + `Send` + `Retry` button	• Send text message • Tap `Retry` on failed messages • Open linked report or Rx note	• **Loading:** Loading chat thread • **Empty:** `Describe your symptoms` • **Offline:** `Saved Locally` in outbox • **Pending:** `Sending 0.6 KB...` • **Success:** `Synced 10:32 AM` • **Error:** `! Failed — Tap Retry`
P-07 Report Upload & Records	Compress report photos in browser, view OCR table & read doctor notes.	• Camera/file picker • **Canvas Compression Badge (`3.2MB -> 38KB`)** • OCR table (`Pending Doctor Review` / `Verified`) • Signed Consultation Notes list	• Select & compress report image • Upload compressed file • Read OCR values & signed clinical notes	• **Loading:** `Compressing on device...` • **Empty:** `No reports uploaded yet` • **Offline:** Reads cached records offline • **Pending:** `Uploading 38 KB...` • **Success:** `Uploaded & OCR Extracted` • **Error:** `Upload dropped — Tap Retry`
P-08 AI Health Assistant	Non-diagnostic health literacy, medical term explanations & triage guide.	• **Amber Non-Diagnostic Disclaimer Banner** • Question box + rural topic chips • Structured plain-language explanation • Red-flag alert linking to `P-11 SOS`	• Ask about lab terms or general care • Read when to seek clinic care • Jump to Booking or SOS on red flags	• **Loading:** `Fetching health guide...` • **Empty:** Starter literacy cards • **Offline:** Uses built-in Curated Offline Health Guide • **Pending:** Generating response • **Success:** Explanation + safety notice • **Error:** Local glossary fallback
P-09 Medicine Reminders	View doctor-approved medicine schedule and log Taken/Skipped doses.	• **`Doctor-Approved Only`** header • Morning / Afternoon / Night cards • Dose, meal instruction & duration • `Mark Taken` & `Skip` buttons + log	• Tap `Mark Taken` after dose • Tap `Skip` with optional note • Review 7-day adherence %	• **Loading:** Loading schedule • **Empty:** `Appears after doctor approval` • **Offline:** Queues dose logs locally • **Pending:** `Syncing dose log...` • **Success:** `Taken 09:05 AM [Synced]` • **Error:** `Stored locally — Tap Retry`

Screen	Purpose	Main UI Elements	User Actions	States (Loading / Empty / Offline / Pending / Success / Error)
P-10 QR Medical Profile & Consent	Share patient-selected profile fields via a temporary, revocable QR token.	• **`Zero Raw PHI in QR`** notice • Field-category checkboxes + Consent • Expiry selector (`1h`, `24h`, `7d`) • QR SVG + shortcode + `Revoke` button	• Select fields & grant consent • Generate temporary QR token • Show QR at clinic or revoke early	• **Loading:** `Generating signed token...` • **Empty:** `No active QR token` • **Offline:** Shows cached unexpired QR • **Pending:** Issuing/revoking token • **Success:** Active QR + expiry timer • **Error:** `Expired/Revoked — regenerate`
P-11 Emergency SOS	One-tap emergency dialer, family contact, manual location & nearby PHCs.	• **`No Auto-Dispatch`** rose banner • 1-tap `CALL 108` (`tel:108`) button • Family Emergency Contact dial card • GPS button + **Manual Village Landmark** • Nearby 24x7 PHC directory	• Tap to open phone dialer (`108`) • Call family emergency contact • Enter village landmark or request GPS • View nearest PHC phone/distance	• **Loading:** `Checking GPS...` • **Empty:** Prompts landmark entry • **Offline: Works 100% offline** (`tel:` dialer + cached PHCs) • **Pending:** Geolocation prompt • **Success:** Displays landmark + coords • **Error:** GPS denied -> Manual input

3.2 Doctor Portal Wireframes & Screen Specifications (9 Screens: D-01 to D-09)

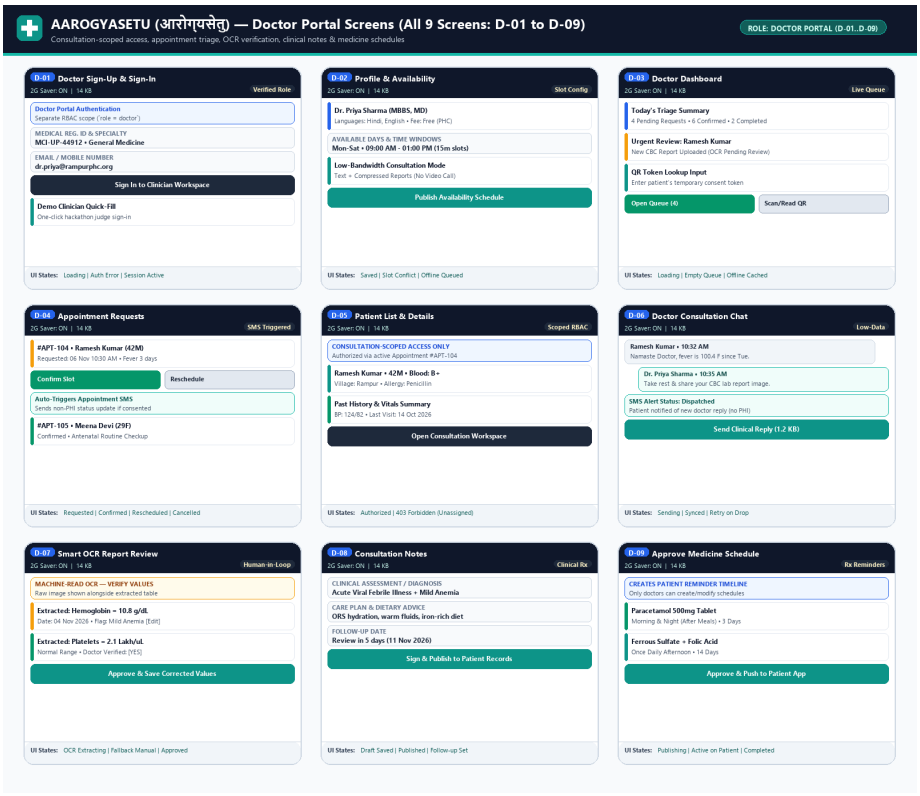


Figure 4: Doctor Portal Mobile & Responsive Wireframes (All 9 Screens: D-01 to D-09).

Screen	Purpose	Main UI Elements	User Actions	States (Loading / Empty / Offline / Pending / Success / Error)
D-01 Doctor Sign-Up & Sign-In	Clinician authentication enforcing `role = doctor` separation.	• Clinician Portal header • Reg. ID, Specialty, Email/Phone • Demo Doctor 1-click sign-in	• Sign in or register as Doctor • Use demo doctor account	• **Loading:** Verifying credentials • **Empty:** Required Reg. ID check • **Offline:** Resumes cached session • **Pending:** Issuing doctor JWT • **Success:** Opens `D-03` Dashboard • **Error:** Auth/role mismatch banner
D-02 Profile & Availability	Configure specialty, spoken languages & consultation slots.	• Quals, Reg. No. & PHC name • Language checkboxes • Weekly day/time slot editor	• Update specialty & languages • Publish open consultation windows	• **Loading:** Loading schedule • **Empty:** Add first time window • **Offline:** Saves draft locally • **Pending:** `Saving schedule...` • **Success:** `Published to directory` • **Error:** Slot overlap warning
D-03 Doctor Dashboard	Daily clinical triage metrics, priority queue & QR token lookup.	• Requested / Confirmed / Done counters • Unreviewed OCR report alerts • **QR Token Lookup input box**	• Open pending requests • Jump to unreviewed reports • Resolve patient QR share token	• **Loading:** Loading queue metrics • **Empty:** `No visits today` • **Offline:** Cached queue + Amber banner • **Pending:** Refreshing queue • **Success:** Live counts updated • **Error:** `Expired/Revoked QR token`
D-04 Appointment Management	Confirm, reschedule, or cancel bookings & trigger patient SMS.	• Status filter tabs • Patient complaint & slot cards • `Confirm`, `Reschedule`, `Cancel` buttons • SMS consent indicator	• Confirm requested slot • Propose rescheduled slot • Cancel with reason	• **Loading:** Loading appointments • **Empty:** `No requests in tab` • **Offline:** Queues status change • **Pending:** `Updating & queuing SMS` • **Success:** `Confirmed + SMS Logged` • **Error:** Update failed — Retry
D-05 Patient List & Details	View assigned patients & consultation-scoped clinical history.	• **`Consultation-Scoped Access`** banner • Assigned patient list • Allergies, conditions, vitals & history	• Select assigned patient • Inspect allergies & past notes • Open consultation workspace	• **Loading:** Loading patient record • **Empty:** `No assigned patients` • **Offline:** Shows cached patient summary • **Pending:** Syncing records • **Success:** Scoped record displayed • **Error:** `403 Forbidden (Unassigned)`

Screen	Purpose	Main UI Elements	User Actions	States (Loading / Empty / Offline / Pending / Success / Error)
D-06 Doctor Consultation Chat	Low-bandwidth clinical messaging with patient SMS reply alert.	• Patient summary strip • Message timeline + sync state • Quick clinical phrase chips	• Send clinical reply • Insert bilingual care phrases • Retry failed messages	• **Loading:** Loading chat thread • **Empty:** `Send first message` • **Offline:** `Saved — Pending Sync` • **Pending:** `Sending 0.9 KB...` • **Success:** `Synced + Reply SMS Queued` • **Error:** `Send failed — Tap Retry`
D-07 Smart OCR Report Review	Verify & edit machine-extracted OCR values side-by-side with image.	• **`Machine-Read — Verify`** banner • Compressed report image viewer • Editable OCR table (Date/Test/Value/Unit) • `Approve & Mark Reviewed` button	• Compare image vs OCR rows • Edit misread values or add rows • Mark report `Doctor Reviewed`	• **Loading:** Running OCR parser • **Empty:** Manual table fallback • **Offline:** Saves review draft locally • **Pending:** `Saving verified table...` • **Success:** `Verified by Doctor` badge • **Error:** Save failed — Retry
D-08 Consultation Notes	Author & sign clinical assessment, care advice & follow-up date.	• Assessment / Diagnosis input • Care & dietary advice box • Follow-up date picker • `Sign & Publish Note` button	• Write structured clinical note • Set follow-up date • Publish to patient records	• **Loading:** Loading note history • **Empty:** Blank clinical template • **Offline:** Auto-saves local draft • **Pending:** `Signing & publishing...` • **Success:** `Published to Patient Record` • **Error:** Draft preserved — Retry
D-09 Approve Medicine Schedule	Doctor-only builder for patient medication reminder schedules.	• **`Doctor-Only Creation`** banner • Drug name, dose, slot & meal inputs • Patient adherence log viewer • `Approve & Push Schedule` button	• Add prescribed medicines & slots • Approve schedule for `P-09` • Review patient adherence %	• **Loading:** Loading schedules • **Empty:** `No active schedule` • **Offline:** Queues draft schedule • **Pending:** `Publishing schedule...` • **Success:** `Active on Patient App` • **Error:** Validation / network retry

SECTION 4 | DETAILED FEATURE SPECIFICATIONS

4.1 Appointment Lifecycle & Appointment Updates Module (SMS Notifications)

• **Booking, Confirmation, Rescheduling & Cancellation:** Patients book open slots (`requested`), reschedule, or cancel. Doctors confirm, reschedule (proposing a new date/time), cancel, or complete consultations. Every change is persisted with an audit timestamp.

• **Appointment Updates Module (SMS):** Automatically handles 6 lifecycle triggers: **Booking Confirmations**, **Scheduled Reminders**, **Doctor Replies**, **Doctor Confirmations**, **Rescheduling**, and **Cancellations**.

• **Consent, Zero-PHI Privacy & Demo Fallback:** Requires `patient.sms_consent == True`. SMS bodies never contain medical details, symptoms, lab results, or drug names—only appointment reference `#APT-xxx`, doctor name, and date/time. Uses a deterministic `dedupe_key` stored in the same database transaction to prevent duplicates, dispatching via Twilio when configured or logging to the clearly labeled **Demo SMS Outbox**.

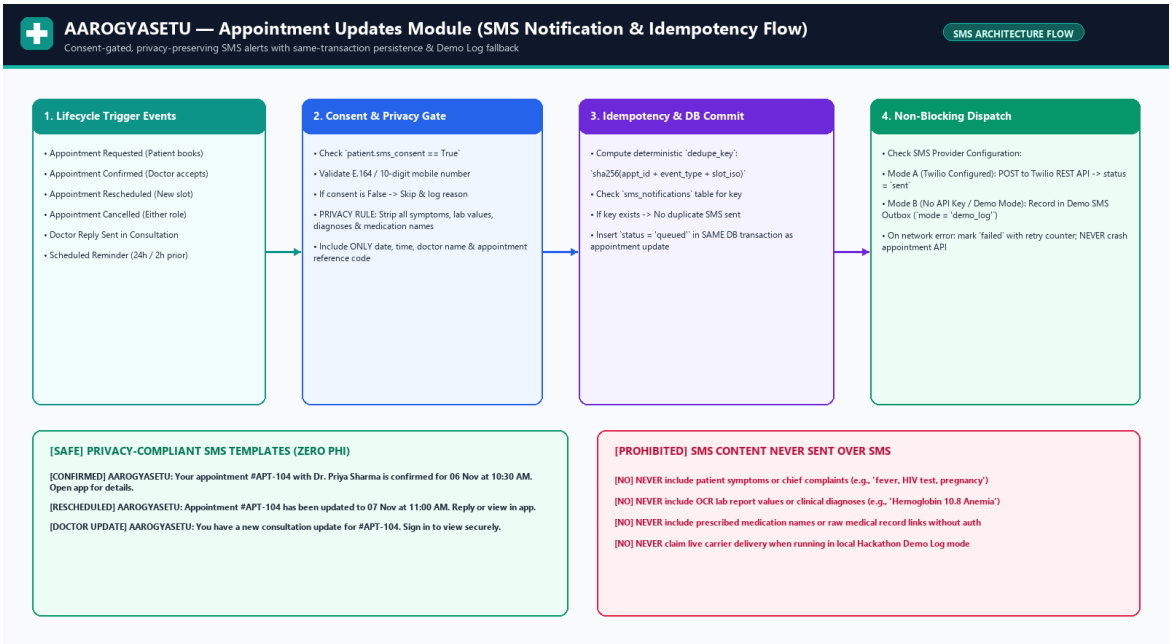


Figure 5: Appointment Updates Module — Consent Gate, Zero-PHI Filter, Idempotent Queue & Demo Fallback Flow.

4.2 Clinical Support, Patient Safety & Low-Bandwidth Modules

Feature Module	Functional & Safety Specification
AI Health Assistant (Non-Diagnostic)	• Explains common medical terms (e.g., Hemoglobin, Platelets, ORS) and general health literacy in plain English/Hindi, plus guidance on when to visit a doctor. • **Hard Guardrail:** Never diagnoses conditions or prescribes medicines. Detects red-flag symptoms (chest pain, breathing distress) and surfaces a direct link to Emergency SOS (`P-11`). Includes an offline curated fallback.
Emergency SOS (No Auto-Dispatch)	• Provides configurable 1-tap phone calling (`tel:108` / local ambulance), saved family emergency contact dialer, optional browser GPS permission prompt, a **Manual Village / Landmark entry box**, and a cached directory of nearby PHCs. • **Hard Guardrail:** Explicitly states that the app does **not** automatically dispatch emergency services; calls are initiated by the user via the device dialer.
QR Medical Profile (Tokenized Sharing)	• Patient selects specific profile categories to share, checks an explicit consent box, and picks an expiration time (`1h`, `24h`, `7d`). • **Hard Guardrail:** The QR code encodes **only a temporary signed token** (`ASETU-TK-...`), never raw medical information. Tokens expire automatically and can be revoked by the patient with one tap.
Doctor-Approved Medicine Reminders	• **Doctor-Only Creation:** Only authenticated doctors (`D-09`) can create or approve a medication schedule (drug name, dosage, Morning/Afternoon/Night timing, meal relation, duration). • Patients view active reminders (`P-09`), mark each dose `Taken` or `Skipped (with note)`, and build an adherence log visible to the doctor during follow-up.
Smart Medical Report Reader (OCR)	• Extracts test dates, parameter labels, numeric values, and units from uploaded lab report photos. • **Hard Guardrail:** All OCR output is labeled **"Machine-Read — May Be Incomplete — Requires Doctor Review"**. Doctors verify and edit extracted values side-by-side with the image (`D-07`) or enter values manually if OCR is unavailable.
Low-Bandwidth & Sync State Engine	• **Client-Side Image Compression:** HTML5 Canvas resizes and compresses `3.2 MB` photos to `~38 KB` JPEG/WebP before upload, displaying real KB saved. • **Explicit Connectivity & Item States:** Header includes a demo-labeled `Simulate Offline` switch and Amber Offline Banner. Every action shows `Saved` -> `Pending` -> `Synced` or `Failed (Tap to Retry)`.

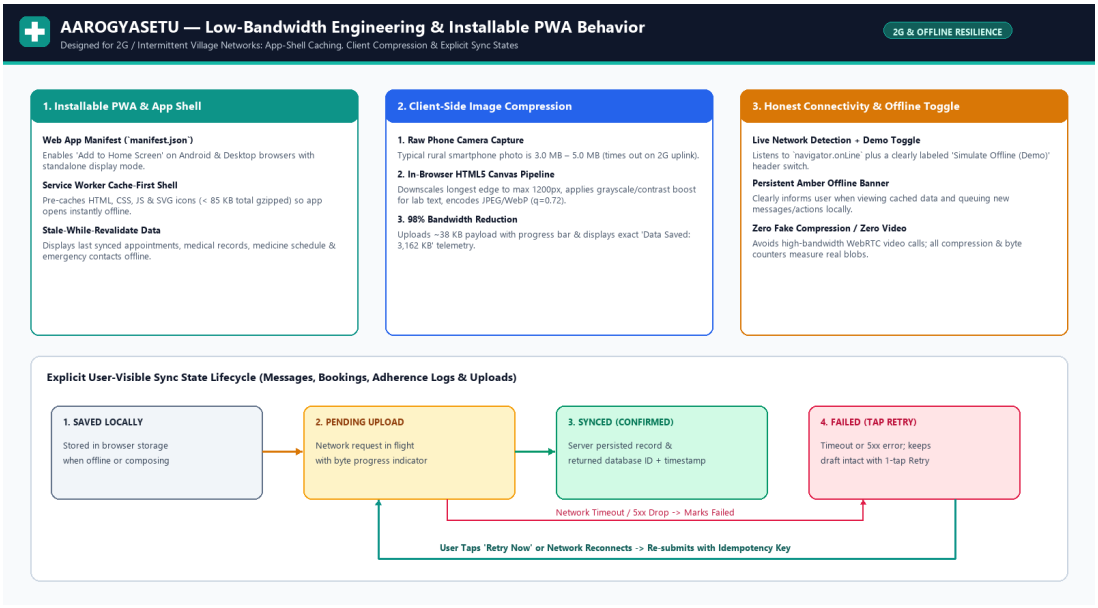


Figure 6: Low-Bandwidth Engineering — Installable PWA App Shell, Client Canvas Compression & Sync Lifecycle.

SECTION 5 | TECHNICAL DEVELOPMENT BLUEPRINT

5.1 Recommended Architecture & System Data Flow

• **Frontend:** React 18 + TypeScript + Vite as an installable Progressive Web App (`manifest.json` + Service Worker `sw.js`).
• **Backend:** Python FastAPI REST API with Pydantic v2 validation and JWT Bearer role authentication.
• **Database & Storage:** SQLite (`aarogyasetu.db`) for zero-config local hackathon demos with a drop-in SQLAlchemy migration path to PostgreSQL; private local file storage (`backend/storage/reports/`) gated by RBAC.
• **Optional Integrations with Demo Fallbacks:** Twilio SMS (`Demo SMS Log` fallback), Gemini/OpenAI (`Offline Curated Health Guide` fallback), and Tesseract OCR (`Structured Parser / Manual Doctor Review` fallback) configured strictly via backend environment variables.

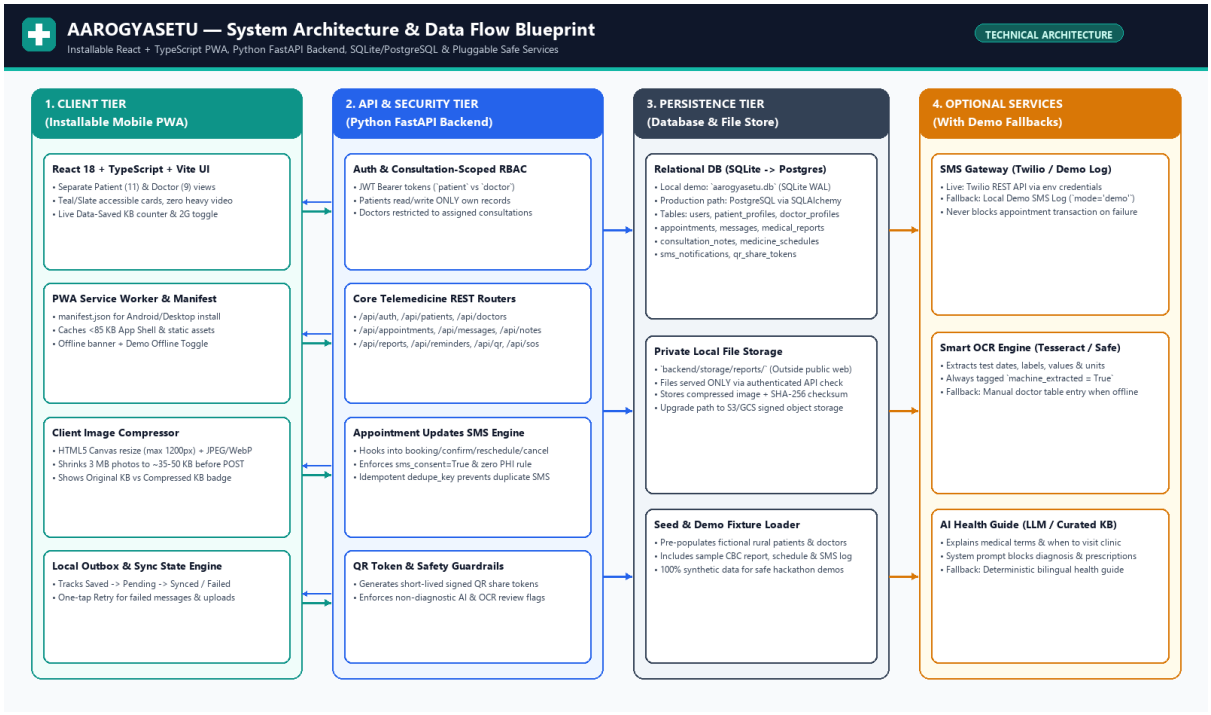


Figure 7: AAROGYASETU 4-Tier System Architecture & Data Flow Diagram.

5.2 Database Entities & Relationships

Table / Entity	Key Columns & Types	Relationships & Access Scope
users	id (PK), phone (Unique), email, password_hash, role ('patient'|'doctor'), full_name, created_at	1:1 with `patient_profiles` or `doctor_profiles`
patient_profiles	id (PK), user_id (FK), age, dob, gender, village_or_town, blood_group, allergies, chronic_conditions, emergency_contact_name/phone, manual_location_landmark, sms_consent (Bool)	1:N with `appointments`, `medical_reports`, `sms_notifications`, `qr_share_tokens`
doctor_profiles	id (PK), user_id (FK), registration_number, specialty, qualifications, languages_json, clinic_name, availability_json, is_accepting	1:N with `appointments`, `consultation_notes`, `medicine_schedules`
appointments	id (PK), reference_code ('APT-104'), patient_id (FK), doctor_id (FK), scheduled_date, scheduled_time, status ('requested'|'confirmed'|'rescheduled'|'cancelled'|'completed'), chief_complaint, status_reason	1:N with `messages`, `medical_reports`, `consultation_notes`, `medicine_schedules`, `sms_notifications`
messages	id (PK), appointment_id (FK), sender_id (FK), sender_role, content, client_msg_id (Unique idempotency key), byte_size, created_at	Belongs to `appointments`
medical_reports	id (PK), appointment_id (FK), patient_id (FK), title, file_path (Private), original_size_bytes, compressed_size_bytes, ocr_status, extracted_fields_json, doctor_reviewed (Bool), doctor_corrected_fields_json, doctor_review_note	Belongs to `appointments` & `patient_profiles`; reviewed by `doctor_profiles`
consultation_notes	id (PK), appointment_id (FK), patient_id (FK), doctor_id (FK), assessment, care_advice, red_flag_warnings, follow_up_date, created_at	Authored by assigned Doctor for Patient
medicine_schedules & adherence_logs	schedule: id, appointment_id, patient_id, doctor_id, medicine_name, dosage_strength, schedule_times_json, meal_instruction, duration_days log: id, schedule_id (FK), dose_date, dose_slot, status ('taken'|'skipped'), note	Created only by Doctor; adherence logged by Patient
sms_notifications	id (PK), appointment_id (FK), patient_id (FK), recipient_phone, event_type, message_body (Zero PHI), dedupe_key (Unique), delivery_mode ('live_twilio'|'demo_log'), status	Created in same transaction as appointment/message event
qr_share_tokens	id (PK), patient_id (FK), token (Unique random string), shared_scopes_json, consent_confirmed (Bool), expires_at, is_revoked (Bool), access_count	Generated/revoked by Patient; resolved by Doctor

5.3 REST API Endpoint Plan & Role-Based Access Control (RBAC)

REST Endpoints	RBAC Enforcement	Functional Summary
POST /api/auth/register **POST /api/auth/login** **GET /api/auth/me**	Public / Authenticated	Register patient or doctor account, issue role-scoped JWT (`patient` vs `doctor`), and fetch current session profile.
GET, PUT /api/patients/me/profile	Patient Only (Own ID)	Read or update the patient's own demographics, allergies, emergency contact, manual landmark, and `sms_consent` toggle.
GET /api/doctors **PUT /api/doctors/me/profile**	Patient & Doctor / Doctor Only	Search doctors by specialty & language (`GET`); update doctor qualifications and availability slots (`PUT`).
POST /api/appointments **GET /api/appointments** **PATCH /api/appointments/{id}/status**	Patient (Book) / Scoped Patient & Doctor	Create booking (`requested`), list own/assigned appointments, and transition status (`confirmed`, `rescheduled`, `cancelled`, `completed`) with automatic SMS queueing.
GET /api/appointments/{id}/sms-logs	Scoped Patient & Doctor	View consent-gated, zero-PHI SMS notification audit trail (`live_twilio` vs `demo_log`).
GET, POST /api/appointments/{id}/messages	Scoped Patient & Doctor	Fetch consultation thread or send low-bandwidth text message (idempotent via `client_msg_id`; doctor reply queues SMS).
POST /api/reports/upload **GET /api/reports/{id}/file** **PUT /api/reports/{id}/review**	Patient (Upload) / Scoped Patient & Doc / Scoped Doctor (Review)	Upload client-compressed lab image & run OCR; stream private file after RBAC check; doctor verifies/edits OCR table and sets `doctor_reviewed = True`.
POST /api/appointments/{id}/notes **GET /api/records**	Scoped Doctor (Create) / Scoped Patient & Doctor	Doctor signs & publishes clinical consultation note; patient and assigned doctor view chronological medical records.
POST /api/reminders/schedules **GET /api/reminders/schedules** **POST /api/reminders/schedules/{id}/log**	Scoped Doctor (Create) / Scoped Patient & Doc / Patient (Log Dose)	Doctor-only creation of approved medicine schedules; patient marks daily doses `taken` or `skipped`.
POST /api/qr/generate **POST /api/qr/{token}/revoke** **GET /api/qr/resolve/{token}**	Patient (Gen/Revoke) / Doctor (Resolve)	Generate or revoke temporary tokenized QR profile; resolve consented fields if token is unexpired and unrevoked.
POST /api/assistant/ask **GET /api/sos/info**	Patient	Non-diagnostic health literacy Q&A; with red-flag triage alert; retrieve emergency contact, `108` helpline & nearby PHCs.

5.4 PWA Installation, Offline Behavior & Hackathon Demo Plan

• **Installable PWA Architecture:** `public/manifest.json` declares standalone display, theme color `#0f172a`, and home-screen icons. Service Worker `public/sw.js` pre-caches the `< 85 KB` App Shell (`index.html`, CSS/JS, icons) on first visit and caches read-only GET responses (`doctors`, `records`, `reminders`, `sos`) for offline viewing.

• **3-Minute Hackathon Demo Plan:** (1) Sign in as rural patient Ramesh Kumar, show PWA installability, enable SMS consent, book Dr. Priya Sharma, and upload a `3.2 MB` report compressed client-side to `38 KB`. (2) Toggle `Simulate Offline (Demo)` to show the Amber Offline Banner and queued message state (`Saved -> Pending -> Synced`), then ask the AI Health Assistant about Hemoglobin. (3) Sign into the Doctor Portal as Dr. Priya Sharma, confirm the appointment (triggering the zero-PHI SMS log), verify the OCR report values, sign a clinical note, and approve a Medicine Schedule. (4) Return to Patient Portal to mark a medicine dose `Taken`, generate a 24h Tokenized QR code, and preview Emergency SOS.

SECTION 6 | PRIVACY, SECURITY AND CLINICAL SAFETY COMMITMENTS

Safety & Privacy Pillar	Mandatory Implementation Rule
1. Strict Patient Data Isolation	Patients can access only their own profile, appointments, messages, reports, medicine reminders, and QR tokens.
2. Consultation-Scoped Doctor Access	Doctors can access patient data only for patients assigned to their consultations (or via an active, unexpired patient-consented QR token).
3. Consent-Gated, Zero-PHI SMS	SMS notifications require explicit patient consent (`sms_consent = True`) and never contain medical details, symptoms, diagnoses, or drug names.
4. Tokenized QR Codes Only	QR codes encode a secure temporary token (`ASETU-TK-...`) with expiration and 1-tap revocation—never raw medical information directly.
5. Non-Diagnostic AI Assistant	The AI Health Assistant provides basic health education, medical term definitions, and guidance on when to seek care; it never diagnoses or prescribes.

Safety & Privacy Pillar	Mandatory Implementation Rule
6. Mandatory Doctor Review of OCR	Machine-extracted OCR lab values may be incomplete or inaccurate and are prominently labeled as requiring doctor verification before clinical use.
7. No Automatic Emergency Dispatch	Emergency SOS provides 1-tap phone dialer links (`tel:108`), emergency contacts, manual/GPS location notes, and nearby clinics; it never auto-dispatches.
8. Prototype & Fictional Data Notice	This blueprint and prototype are not production-ready certified medical software and must be demonstrated exclusively using fictional demo data.

SECTION 7 | EXPECTED OUTCOME

"A working prototype that demonstrates how technology can make basic healthcare services more accessible to people in rural and remote areas."